

Assignment 4 - Full-stack application

Instructions

In this assignment, you are required to build a full-stack quiz application using Node.js for the backend and React with Redux for the frontend. The backend will use Express and MongoDB to handle data storage and API endpoints, while the frontend will provide an interactive interface for users to take quizzes.

Assignment Overview

At the end of this assignment, you should have completed the following:

- The backend handles the CRUD operations for quizzes and questions
- The frontend allows user login successfully before fetching and displaying quizzes using Redux for state management.

Assignment Requirements

1. Backend

- Express Server: Set up an Express server to handle API requests.
- MongoDB: Use MongoDB for data storage.
- Mongoose Models: Define Mongoose models for Quizzes and Questions.
- Routes: Create API endpoints for creating, reading, updating, and deleting quizzes and questions.
- Validation and Error Handling: Implement validation and error handling for API requests.

2. Frontend

- React: Build the frontend using React.
- Redux: Use Redux for state management.
- Routing: Implement client-side routing with React Router.
- Quiz Display: Display quizzes and their questions.

- Authentication: Login, logout and signup user account. If user is admin, user can CRUD question and other ones can only take quizzes
- User Interaction: Allow users to select options for questions and submit answers.
- Styling: Use Bootstrap 5 to style the application.

Review criteria

Your assignment will be graded on the basis of the following review criteria:

- Have a full-stack Node.js quiz application with an Express backend and a React frontend.
- The backend handles the CRUD operations for quizzes and questions, while the frontend allows user login successfully before fetching and displaying quizzes with Redux architecture. This is a basic setup that can be extended with more features such as user authentication, score tracking, and more.



Login

Username

Password

[Login](#)

[Don't have an account? Register here](#)

Login successfully with user that is not admin

Dashboard

Welcome, John

[Home](#) [Quiz](#) [Article](#) [Logout](#)

Click and do Quiz

Dashboard

Welcome, John

[Home](#) [Quiz](#) [Article](#) [Logout](#)

Quiz

What is the capital of Germany?

- ☐ Paris
- ☐ London
- ☒ Berlin
- ☐ Madrid

Submit Answer

Finish quiz

Dashboard

Welcome, John

[Home](#) [Quiz](#) [Article](#) [Logout](#)

Quiz Completed

Your score: 4

[Restart Quiz](#)

Login

Username

Mary

Password

...

[Login](#)

[Don't have an account? Register here](#)

Login successfully with user that is an admin

Admin Dashboard

Welcome, Mary

Home Manage Questions Manage Articles Logout

Questions

Question Text:

Options:

Correct Answer Index:

0

Add Question

Who developed the theory of relativity?

- Albert Einstein
- Isaac Newton
- Galileo Galilei
- Nikola Tesla

EditDelete

What is the symbol for iron on the periodic table?

- Fe
- I
- Au
- Ag

EditDelete

What is the capital of France?

- Paris
- London
- Berlin
- Madrid

EditDelete

What is the capital of England?

- Paris
- London
- Berlin
- Madrid

EditDelete

What is the capital of Germany?

- Paris
- London
- Berlin
- Madrid

EditDelete

